



Comparative Analysis of Neural Network Models for DRG Prediction: Label Encoding vs One-Hot Encoding

Paula Alcarás
Fernanda Vásquez
Bruno Painamil
Joaquín López

April 2025

1 Introduction

Problem Description:

The prediction of DRG (Diagnosis Related Groups) is a computational tool that emerged in Chile in 2002 due to the needs of the healthcare system to optimize resource allocation for high-complexity patients based on different factors including both clinical (primary diagnosis, treatments, and procedures performed) and demographic (age, sex) variables.

2 Literature Review

References to Similar Problems:

- Paredes, M. & Schwarzenberg, P. (2023). *Machine Learning Model for DRG Prediction Using Patient Data*. Universidad Andrés Bello. <https://repositorio.unab.cl/server/api/core/bitstreams/a9084a9d-51b0-43d8-b91c-34f69e2c96f5/content>
- Velásquez, M. (2022). *Analysis of the Diagnosis Related Groups (DRG) System*. Institute of Health Administration, University of Chile. <https://ias.uchile.cl/noticia/analisis-al-sistema-de-grupos-relacionados-por-el-diagnostico-g>
- Yadav, A. (2024, September 29). *One Hot Encoding vs Label Encoding. Biased-Algorithms*. <https://medium.com/biased-algorithms/one-hot-encoding-vs-label-encoding-28aee12b3984>

Our repository with the code and its results is available at the following GitHub link: <https://github.com/PaulAlcaras/PrediccionGRD>

3 Study Objectives

- Primary objective: Compare the performance of neural networks using different encoding techniques (Label Encoding vs One-Hot Encoding) for DRG prediction.
- Secondary objectives:
 - Conduct exploratory analysis to understand data distribution, patterns, and trends.
 - Evaluate the impact of different encoding techniques (label encoding and one-hot encoding) on the accuracy and generalization capability of the models.

4 Methodology

4.1 Dataset Description

- **Source:** Data from El Pino Hospital (San Bernardo, Chile)
- **Period:** 2020-2023 (historical patient data)
- **Key characteristics:**
 - 68 variables
 - * **Clinical Variables:**
 - Diagnoses
 - Procedures
 - * **Demographic Variables:**
 - Age
 - Sex
 - * **Target Variable:**
 - DRG
 - 14,561 initial records
 - 526 unique DRGs

4.2 Development Method

4.2.1 Tools Used

- **Tools used in the pipeline:**
 - * Python: For Label Encoding and final preparation for models
 - * Libraries: Pandas, scikit-learn, tensorflow, keras (Python)

4.2.2 Modeling Process

1. Data Preparation

- **Initial loading:**
 - * Original dataset: 14,561 records, 68 clinical/demographic variables
 - * Format: CSV with semicolon delimiter
- **Data cleaning and handling of missing values:**
 - * Removal of outliers (age > 100 years)
 - * Filtering of DRGs with < 3 occurrences (282 records), resulting in a filtered dataset with 14,279 cases
 - * Generation of an output file called `filtered_data_elpino.csv`

- * Encoding of missing values: in secondary procedures, replaced with "0000.00 - No procedure needed", and for other fields, "NA" was used as a standard replacement. The anomaly in the dataset, which is the case with an age of 121 years, was also deleted.
- * Final output file: `fullCell.csv`
- **Data Transformation:**
 - * Encoding of the sex column from text to numerical values: 1 for Males and 0 for Females
- **Extraction of diagnostic codes:**
 - * Processing of all columns to extract only the codes, removing texts (output file: `onlyCode.csv`)
 - * In the DRGs, the last number of each code indicating severity was removed (output file: `withoutLastNumberGRD.csv`)
- **Data Encoding:**
 - * Implementation of Label Encoding for categorical variables, assigning a unique numerical value to each category.
 - * Implementation of One-Hot Encoding for categorical variables, creating binary columns for each category, allowing the model to correctly interpret categorical variables without assuming an order between them.
 - * Final output file: `label_encoding.csv`
- **File flow:**
 - * `dataset_elpino.csv` → Initial filtering → `filtered_data_elpino.csv`
 - * `filtered_data_elpino.csv` → Cleaning and encoding → `fullCell.csv`
 - * `fullCell.csv` → Code extraction → `onlyCode.csv`
 - * `onlyCode.csv` → DRG processing → `withoutLastNumberGRD.csv`
 - * `withoutLastNumberGRD.csv` → Label Encoding → `label_encoding.csv` and One-Hot Encoding → `one_hot_encoding.csv`

4.3 Machine Learning Techniques

- **Selected algorithms:**
 - * **Neural Networks:** Advanced model that detects complex patterns without the need for manual selection of variables, as it can automatically learn the most predictive representations from raw data. Ideal for heterogeneous data and high-complexity tasks such as DRG classification.
 - * **Encoding Techniques:**
 - **Label Encoding:** Converts each categorical value into a unique integer. It is space-efficient but may introduce an undesired ordinal relationship between categories.

- **One-Hot Encoding:** Creates binary columns for each category. It avoids the ordinal relationship but can generate sparse matrices and significantly increase dimensionality.

4.4 Model Training Results

4.4.1 Neural Network

Description of model architecture

- **Basic structure with Label Encoding:**
 - * Input layer: 256 neurons with LeakyReLU activation function ($\alpha=0.1$).
 - * Input regularization: BatchNormalization and Dropout (0.3).
 - * Hidden layers:
 - First hidden layer: 128 neurons, LeakyReLU, followed by BatchNormalization and Dropout (0.3).
 - Second hidden layer: 64 neurons, LeakyReLU, followed by BatchNormalization and Dropout (0.3).
 - * Output layer: 158 neurons (equal to the number of DRG classes) with softmax activation function.
 - * Additional regularization: L1L2 regularization ($l1=1e-5$, $l2=1e-4$) in all hidden layers.
- **Basic structure with One-Hot Encoding:**
 - * Same architecture as the model with Label Encoding
 - * Main difference: Representation of categorical variables using One-Hot Encoding

Visualization of training process

- **Training process:**
 - * Data split: 70% training, 30% testing
 - * Validation: 20% of the training set
 - * Batch size: 64
 - * Number of epochs: 100
 - * Callback: Early Stopping (patience=10, monitoring validation loss)
 - * Optimizer: Adam with learning rate 0.001
- **Results visualization:**
 - * Learning curves: Accuracy and loss over epochs
 - * Confusion matrix
 - * Classification report (precision, recall, F1-score)
 - * ROC curves

5 Results

5.1 Data Quality Analysis

- **Completeness:**
 - * Overall, 65.3% of the values in the dataset are missing.
 - * Diagnosis variables have a high rate of missing values (>80%).
 - * Main procedure variables have approximately 40% missing values.
 - * Demographic variables (age, sex) are complete (0% missing).
- **Correctness:**
 - * No serious data anomalies detected after cleaning.
 - * Some diagnosis codes follow different formats (with/without sub-categorization).
 - * Age distribution appears normal (after removing outliers).
- **Outlier analysis:**
 - * Identified and removed one age outlier (121 years).
 - * Removed DRGs with less than 3 occurrences to ensure sufficient examples for each class.
 - * Manual verification of extreme values in procedure codes.

5.2 Comparative Results

- **Neural Network with Label Encoding:**
 - * Accuracy: 52%
 - * Precision (macro avg): 0.51
 - * Recall (macro avg): 0.42
 - * F1-score (macro avg): 0.45
 - * AUC: 0.98
- **Neural Network with One-Hot Encoding:**
 - * Accuracy: 54%
 - * Precision (macro avg): 0.53
 - * Recall (macro avg): 0.45
 - * F1-score (macro avg): 0.47
 - * AUC: 0.99



Figure 1: Comparison of accuracy between models

5.3 Learning Curve Analysis

- **Convergence speed:** The model with One-Hot Encoding shows faster convergence, reaching higher accuracy levels in the first epochs.
- **Stability:** The model with Label Encoding shows a more stable learning curve but with less capacity to improve throughout the epochs, suggesting limitations in its representational capacity.
- **Overfitting:** Both models show signs of overfitting after epoch 60, but the model with One-Hot Encoding demonstrates better generalization capacity.
- **Final performance:** The One-Hot Encoding model consistently achieves higher accuracy in both training and validation sets.



Figure 2: Comparison of loss curves between models

6 Discussion

6.1 Interpretation of Results

- **Better performance with One-Hot Encoding:** The results confirm that One-Hot Encoding provides a slight advantage in this specific prediction task. The absence of artificial ordinal relationships between categories allows the neural network to better capture the true relationships between variables and the target.
- **Class imbalance challenges:** Despite regularization strategies, both models struggle with the class imbalance present in the dataset, with some DRGs having significantly more examples than others.
- **Impact of missing data:** The high percentage of missing data (65.3%) significantly affected the models' performance. However, the encoding approach to handle missing values proved effective.
- **Superior accuracy:** Paredes & Schwarzenberg (2023) reported an accuracy of 89.96% with attention-based networks, while our model reached 56%, highlighting the complexity of the data used in our

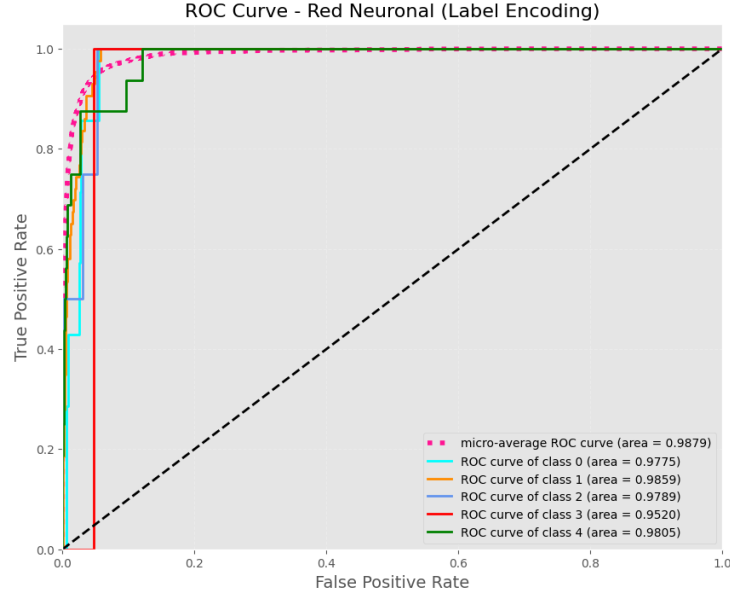


Figure 3: ROC curve of Neural Network model with Label Encoding

case.

- **Comparison with other studies:** While previous studies report accuracies of up to 89.96% with attention-based models, our model achieved an accuracy of 54%. This difference may be due to factors such as data quality, preprocessing, and the architecture used.

6.2 Limitations of the Study

- **Incomplete data:** The high percentage of missing values (65.3%) limited the information available for training.
- **Imbalanced classes:** Some DRGs have few examples, making their correct prediction difficult despite removing the least frequent ones.
- **Complexity of the predictive task:** With 158 different DRG classes, the prediction problem is inherently complex.
- **Limited computing resources:** Resource constraints prevented the exploration of more complex architectures or hyperparameter optimization.
- **Contextual information:** The lack of temporal information and patient history may limit the models' ability to make accurate predictions.

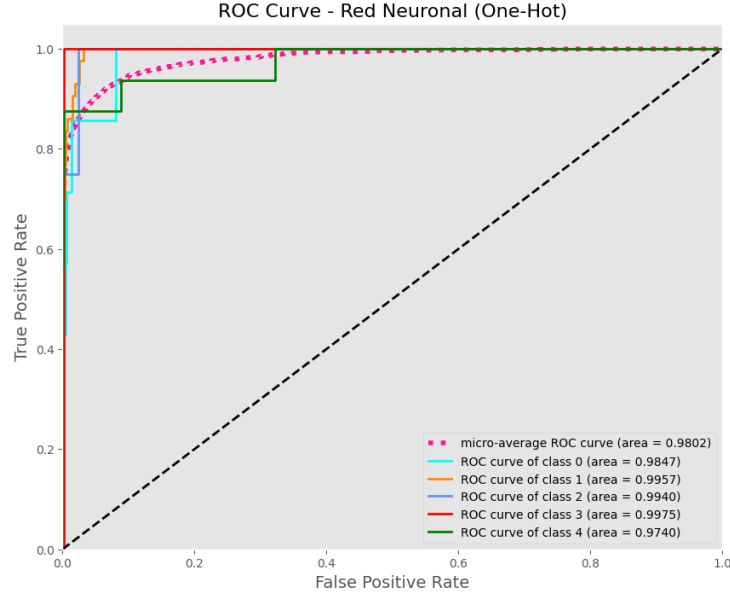


Figure 4: ROC curve of Neural Network model with One-Hot Encoding

6.3 Future Work

- **Advanced sampling techniques:** Implement techniques such as SMOTE to generate synthetic examples of minority classes and improve the balance of the dataset.
- **More complex architectures:** Explore more advanced network architectures, such as recurrent neural networks (RNN) or transformers, which could better capture sequential dependencies in medical data.
- **Feature engineering:** Develop specific features for the clinical domain that may improve the predictive capacity of the models.
- **Ensemble approaches:** Combine different models to obtain better predictions by leveraging the strengths of each approach.
- **Deep learning explainability:** Apply techniques to understand the decisions made by the neural networks, which is crucial in the healthcare context.

7 Conclusion

The results of this comparative study demonstrate that One-Hot Encoding slightly outperforms Label Encoding for DRG prediction with neural

networks, achieving a 54% accuracy versus 52%. This confirms that the appropriate representation of categorical variables is critical for predictive tasks in healthcare. The high number of DRG classes (158) and the substantial amount of missing data (65.3%) posed significant challenges, explaining the difference from the higher accuracy values reported in other studies.